

# Discrete Mathematics

## Graph Theory



# Introductions

Graph theory is an important area of study in Mathematics which enables us to solve a large variety of problems in an easy and simple manner.

Many situations that occur in the fields of Computer Science, Physical Science, Electrical Engineering, and many other areas can be analyzed by using techniques described in graph theory. Graphs can be used for solving the following types of problems:

- (i) What is the shortest route from the national capital to every state capital?
- (ii) What is the maximum flow per unit time from source to sink in a network of pipes?
- (iii) How many layers does a computer chip need so that wires in the same layer do not cross?
- (iv) In what order should a travelling salesperson visit cities to minimize his/her travel time?

# Definitions-Graph

- **Graph:**

A graph  $G=(V,E)$  consists of  $V$ , a nonempty set of vertices (or nodes or points), and  $E$ , a set of edges (or lines) such that each edge  $e \in E$  is associated with ordered or unordered pair of elements of  $V$ .

- (ii) Mapping from the set of edges  $E$  to the set of ordered or unordered pairs of elements of  $V$ .

**Note:**

A graph  $G_1$  is an ordered triple  $(V, E, \phi)$

where

→  $V$  = set of vertices

→  $E$  = set of edges

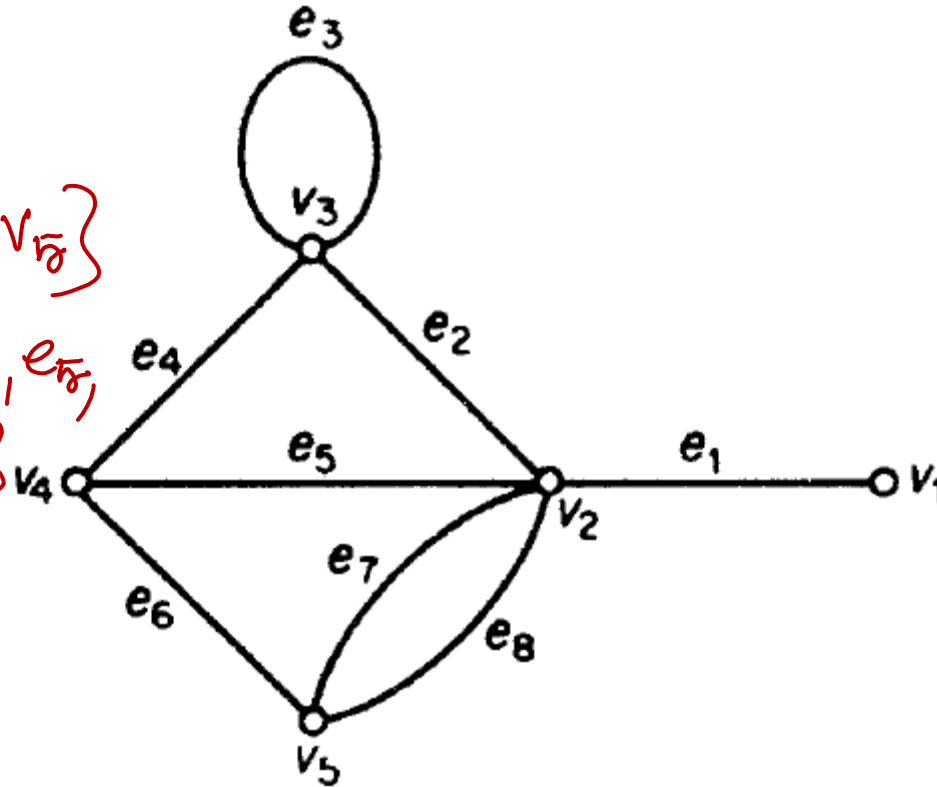
→  $\phi$  = mapping from the set of edges  $E$  to a set of ordered or unordered pairs of elements of  $V$ .

diagram of  $G$

$$G = (V, E)$$

$$V = \{v_1, v_2, v_3, v_4, v_5\}$$

$$E = \{e_1, e_2, e_3, e_4, e_5, e_6, e_7, e_8\}$$

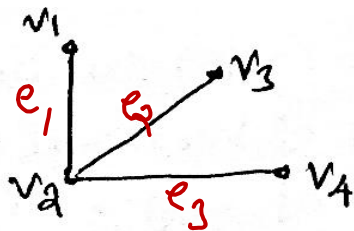


## Finite Graph

A graph  $G_1 = (V, E)$  is said to be **finite** if it has a finite number of vertices and a finite number of edges. Otherwise, it is called an **infinite graph**.

## Adjacent Vertices

Any pair of vertices which are connected by an edge in a graph is called **adjacent vertices**.



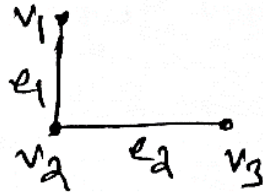
here

$v_1, v_2$ ;  $v_2, v_3$ ;  $v_2, v_4$  are adjacent vertices

$v_1, v_3$ ;  $v_3, v_4$ ;  $v_1, v_4$  are not adjacent

## Adjacent edges

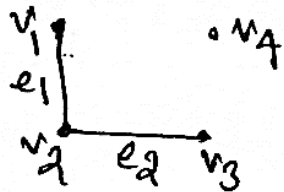
If two distinct edges are incident with a common vertex then they are called adjacent edges.



here  $e_1$  and  $e_2$  are incident with a common vertex  $v_2$

## Isolated vertex

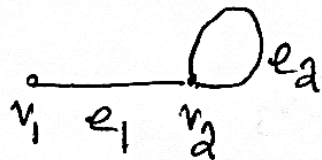
In any graph, a vertex which is not adjacent to any other vertex is called an isolated vertex.



here  $v_4$  is isolated vertex

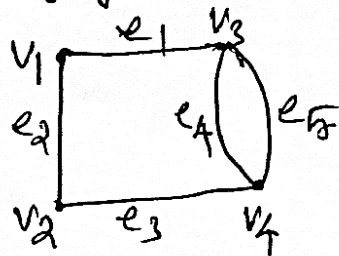
## Loop

An edge of a graph that joins a vertex to itself is called a loop.



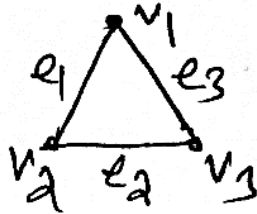
## Multi graph

Any graph which contains some parallel edges is called as multi graph.



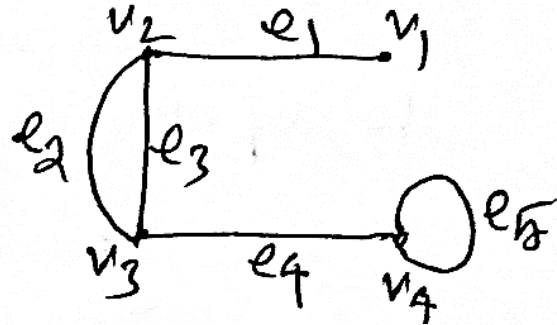
## Simple graph

A graph which has neither loops nor multiple edges is called a simple graph



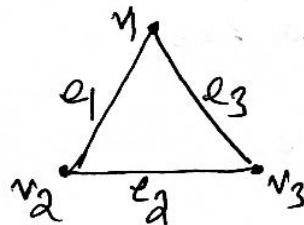
## Pseudograph

A graph in which loops and multiple edges are allowed is called a pseudograph



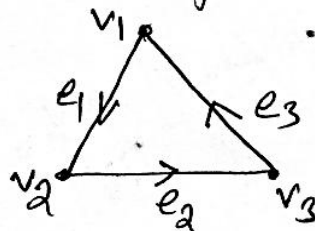
## undirected graph

An undirected graph  $G$  consists of a set  $V$  of vertices and a set  $E$  of edges such that each edge  $e \in E$  is associated with an unordered pair of vertices



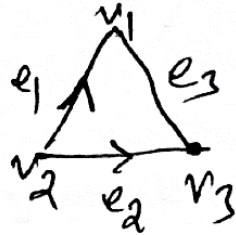
## Directed graph

If each edge of the graph  $G$  has a direction then the graph is called directed graph



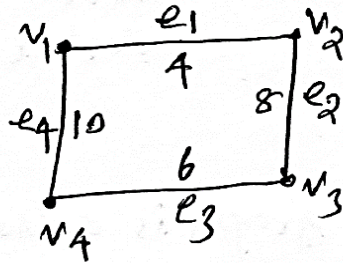
## Mixed graph

If some edges are directed and some are undirected in a graph then the graph is called mixed graph



## Weighted graph

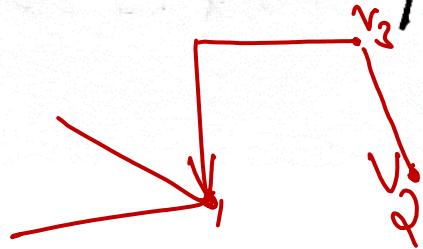
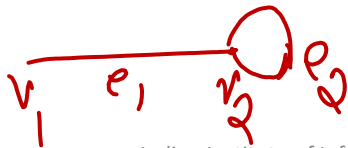
A graph in which weights are assigned to every edge is called a weighted graph



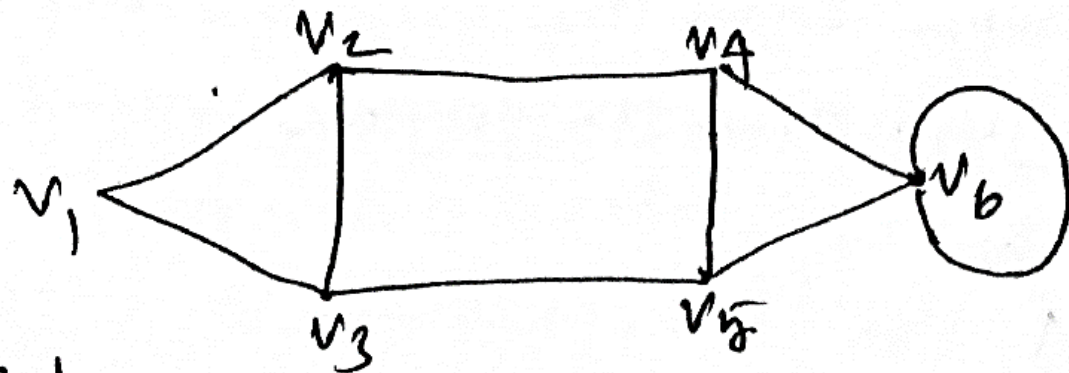
here 4, 8, 6, 10 are weights assigned to each edge respectively.

## Degree of a vertex

The degree of a vertex of an undirected graph is the number of edges incident with it, except that a loop at a vertex contributes twice to the degree of that vertex. The degree of the vertex  $v_i$  in a graph  $G$  is denoted by  $\deg_G(v_i)$  or  $\deg v_i$  or  $d(v_i)$



Find the degree of each vertex of the graph



Soln

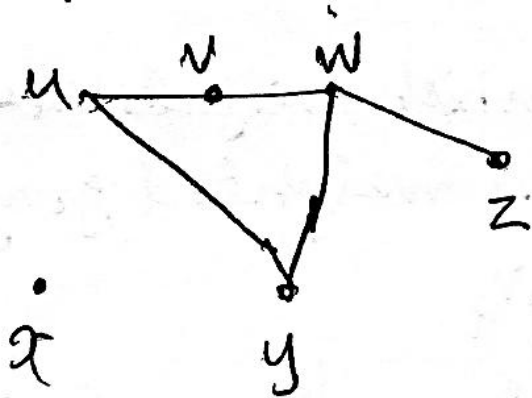
$$\deg(v_1) = 2, \deg(v_2) = 3, \deg(v_3) = 3$$
$$\deg(v_4) = 3, \deg(v_5) = 3, \deg(v_6) = 4$$

## Note

A vertex of degree zero is called an isolated vertex.

A vertex of degree one is called an end vertex.

A vertex is pendant if and only if it has a degree 1.



here  $x$  is an isolated vertex and the vertex  $z$  is pendant

## odd and even vertices

A vertex of a graph is called an odd vertex or an even vertex depending on ~~it~~ whether its degree is odd or even.

## In degree of a vertex

The in-degree of a vertex  $v$  of a directed graph  $G$  is the number of edges ending at  $v$  and is denoted by  $\text{indeg}_G(v)$  or  $\text{deg}_G^-(v)$ .

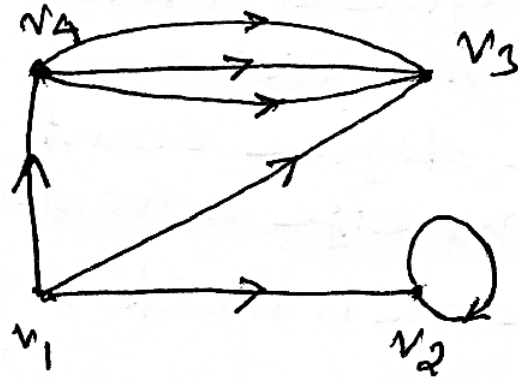


out-degree of a vertex  $v$

The out-degree of a vertex  $v$  of a directed graph  $G$  is the number of edges beginning at  $v$  and is denoted by  $\text{outdeg}_G(v)$  or  $\text{deg}_G^+(v)$

Sum of the in-degree and out-degree of a vertex is called the total degree of the vertex.

Find the in degree, out-degree and total degree of each vertex of the graph



Soln

$$\text{in deg}(v_1) = 0$$

$$\text{in deg}(v_2) = 2$$

$$\text{in deg}(v_3) = 4$$

$$\text{in deg}(v_4) = 1$$

$$\text{out degree}(v_1) = 3$$

$$\text{out degree}(v_2) = 1$$

$$\text{out degree}(v_3) = 0$$

$$\text{out degree}(v_4) = 3$$

$$\text{total deg}(v_1) = 3$$

$$\text{total deg}(v_2) = 3$$

$$\text{total deg}(v_3) = 4$$

$$\text{total deg}(v_4) = 4$$

Let  $e$  denote the number of edges of a graph  $G$  with  $n$  vertices  $v_1, v_2, \dots, v_n$ .

Then  $\sum_{i=1}^n \deg(v_i) = 2e$ .

i.e., Sum of degree of the vertex in a graph =  $2 \times (\text{no of edges})$

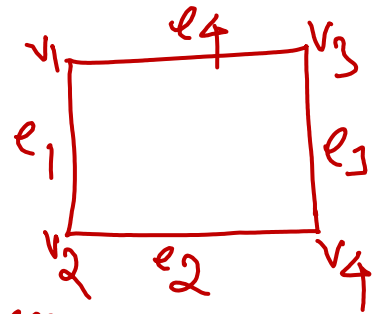
proof-

let  $G = (V, E)$  be any graph

where  $V = \{v_1, v_2, \dots, v_n\}$

$E = \{e_1, e_2, \dots, e_n\}$

Since degree of a vertex is no of edges incident with a vertex



here  
 $\deg v_1 = \deg v_2 = \deg v_3$

$= \deg v_4 = 2$

no of edges = 4

$2+2+2+2 = 2 \times 4$

$8 = 8$

Sum of degree count is no of times edges  
incident with a vertex

here edge is connection between exactly  
two vertices

i.e. every edge is counted twice

$\therefore$  Sum of degree of the vertices in  $G$   
is equal to twice the number of edges

$$\sum_{i=1}^n \deg(v_i) = 2e$$

In any graph  $G$ , the total number of odd-degree vertices is even.

proof → let  $G$  be any graph

and let  $U$  be the set of all even degree vertices in  $G$

let  $W$  be the set of all odd degree vertices in  $G$

w.k.t  $\sum_{i=1}^n \deg_G(v_i) = 2e$

$$\Rightarrow \sum_{v_i \in U} \deg_G(v_i) + \sum_{v_i \in W} \deg_G(v_i) = 2e$$

$$\Rightarrow \sum_{v_i \in W} \deg_G(v_i) = 2e - \sum_{v_i \in U} \deg_G(v_i)$$

3, 3, 5, 3

$$\left( \sum_{v_i \in W} \deg_G(v_i) = \text{even} \right) \therefore \text{Hence proved}$$

Show that the degree of a vertex of a simple graph  $G$  on  $n$  vertices cannot exceed  $(n - 1)$ .

Soln

Let  $G$  be the graph with  $n$  vertices

here  $G$  is Simple graph

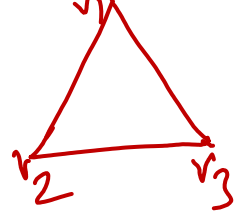
i.e. (not allowed multiple edges and Loops)

now every vertex in  $G$  adjacent to  
remaining  $(n-1)$  vertices

i.e. degree of a vertex cannot exceed  $(n-1)$

i.e. maximum degree is  $(n-1)$  only

Example



Show that the maximum number of edges in a simple graph with  $n$  vertices is  $\frac{n(n-1)}{2}$ .

Soln

Let  $G$  be the Simple graph with  $n$  vertices

$$\text{w.k.T } \sum_{i=1}^n \deg(v_i) = 2e$$

$$\Rightarrow \deg(v_1) + \deg(v_2) + \dots + \deg(v_n) = 2e$$

w.k.T in a Simple graph with  $n$  vertices  
has maximum degree  $(n-1)$

$$(n-1) + (n-1) + \dots + (n-1) = 2e$$

$$\Rightarrow \frac{n(n-1)}{2} = e$$

$$\therefore e = \frac{n(n-1)}{2}$$

Is there a simple graph corresponding to the following degree sequences:

(i) (2, 2, 4, 6)

(ii) (1, 1, 2, 3)  $\rightarrow$  (here odd no of degree seq is odd  $\Rightarrow$  not possible)  
ie (1, 1, 3)

Soln (i) Given degree sequence is (2, 2, 4, 6)

ie, A graph with 4 vertices

w.k.T a graph with  $n$  vertices has  
maximum degree is  $(n-1)$

but given sequence  $\therefore$  here maximum degree is 3 only  
~~maximum degree is 6~~

$\therefore$  which is not possible  
There is no graph with given degree sequence



Does there exist a simple graph with seven vertices having degrees  $(1, 3, 3, 4, 5, 6, 6)$ ?



